**MAST10005 Assignment 7**

Semester 2 2023

Due Date: Thursday 5 October

Assignments are due in GradeScope at 11:59pm on the due date listed above. Late assignments are not accepted. Please see the instructions on the page where you accessed this file for details on submitting your assignment in Gradescope.

Students are encouraged to work together on understanding the problems and their solutions. However, submitted solutions must be prepared individually, in your own words, and without the aid of others. Answers presented without justification will receive no marks.

If you do not completely understand your submitted solutions, it is possible that you are committing academic misconduct. This is a serious offense.

Do not upload this assignment paper or any of its questions/solutions to any online *"help"* service (e.g., CourseHero, Chegg, ChatGPT, etc...). By doing so, you are actively ruining the learning experience for yourself and future students. In doing so you are more than likely violating the University's academic misconduct policy.

**Instructions:** Use the boxes provided to write up your final draft of the work you are submitting to be assessed. You will need to scan these pages in order to submit to Gradescope.

You do not need to scan this cover page.

**PART A – WebWork** *(3 marks)*

Complete Part A using WebWork. See *Assignments* on Canvas for an Assignment named *Assignment 7 WebWork part* to complete this question.

There is one WebWork problem.

**See over for PART B**

**Part B.** In this question we will use the answers you obtained in Part A to understand the graph of

$$f(x) = \frac{x^3 + 2x^2 + 9}{x^2 - 9}.$$

(1) (*2 marks*) Use your answer to Part A to find any asymptotes of $f$.

From Webwork, polynomial long division gives

$$f(x) = x + 2 + \frac{9}{x-3}.$$

Thus the function $f$ has a vertical asymptote at $x = 3$ and an oblique asymptote with equation $y = x + 2$.

(2) (*4 marks*) Use your answers to Part A to find the stationary points of $f$ (*remember that these are points on the graph of* $f$). Find the intervals on which $f$ is increasing and the intervals on which $f$ is decreasing.

We have $f'(x) = 1 - \dfrac{9}{(x-3)^2} = \dfrac{x^2 - 6x}{(x-3)^2} = 0 \implies x^2 - 6x = x(x-6) = 0 \implies x \in \{0, 6\}$ giving stationary points at $(0, f(0)) = (0, -1)$ and $(6, f(6)) = (6, 11)$.

We have

$$f'(x) = \frac{x^2 - 6x}{(x-3)^2} > 0 \implies x^2 - 6x > 0 \implies x < 0 \text{ or } x > 6.$$

Thus $f$ is increasing on $(-\infty, -3) \cup (-3, 0) \cup (6, \infty)$. We have

$$f'(x) = \frac{x^2 - 6x}{(x-3)^2} < 0 \implies x^2 - 6x < 0 \implies 0 < x < 6.$$

Thus $f$ is decreasing on $(0, 3) \cup (3, 6)$. (Notice that $-3 \notin \mathrm{dom}(f)$ and $3 \notin \mathrm{dom}(f)$.)

(3) (*3 marks*) Use your answer to Part A to find interval(s) on which $f$ is concave up and the interval(s) on which $f$ is concave down. Determine whether or not the function $f$ has any inflection points. Explain your reasoning.

We have

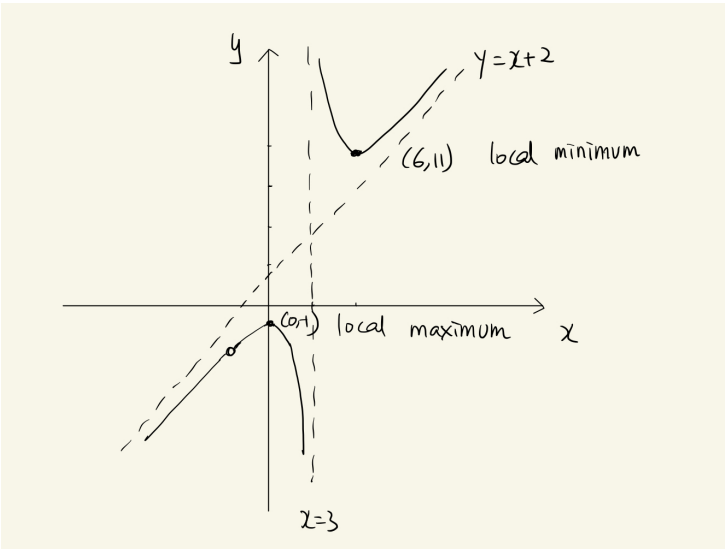
$$f''(x) = 18(x-3)^{-3} > 0 \implies x > 3 \text{ and } f''(x) = 18(x-3)^{-3} < 0 \implies x < 3.$$

Thus $f$ is concave up on the interval $(3, \infty)$, and $f$ is concave down on $(-\infty, -3) \cup (-3, 3)$.

The function $f$ does not have any inflection points since $f$ changes concavity at $x = 3$ but $3 \notin \operatorname{dom} f$.

(4) (*2 marks*) Determine the nature (local maximum, local minimum, or neither) of the stationary point(s) you found in (2). Explain your reasoning.

When $x < 0$ the function $f$ is increasing and when $0 < x < 3$ the function $f$ is decreasing, so $(0, -1)$ is a local maximum.

Similarly, when $3 < x < 6$ the function $f$ is decreasing, and when $x > 6$ the function $f$ is increasing, so $(6, 11)$ is a local minimum.

(5) (*2 marks*) Use your answers to parts (1) – (4) to draw a sketch of the graph of $f$, showing the asymptotes and labelling points you found in the previous parts.